JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI DEVI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

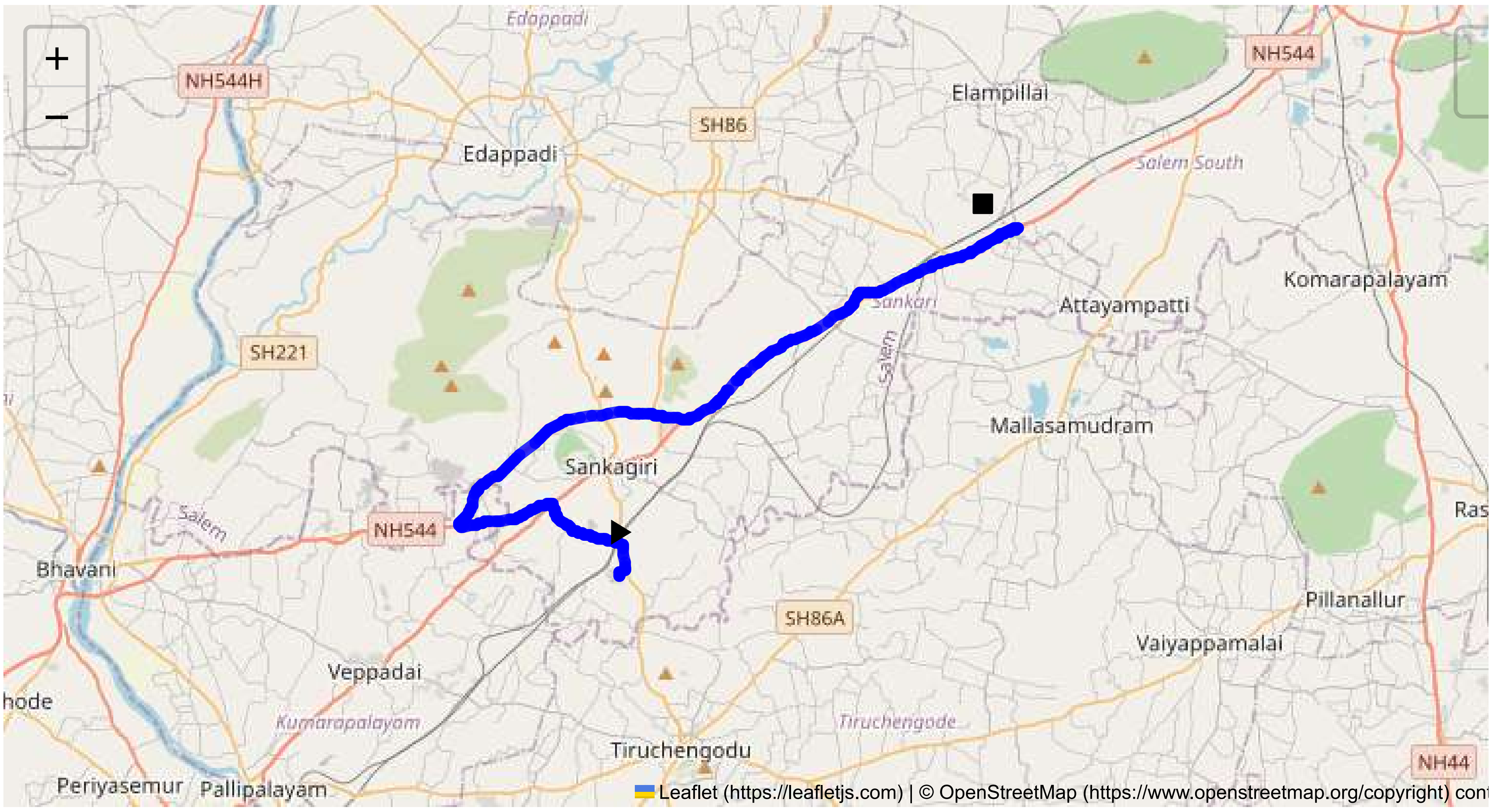
Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 35.70 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.551309, 78.000027)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, to SF No 53/2B2, Kanagagiri, is approximately 35.70 kilometers long. It traverses through several small towns and highways, including Puthur and Padaiveedu, primarily along the Salem - Kochi Highway. This route is moderately trafficked, connecting rural and semi-urban regions with varying road conditions and infrastructural quality.

Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate. The region can have heavy monsoon rains between October and December that may cause road flooding, reducing visibility and making roads slippery. Summer months are typically hot and dry, leading to risks of tire blowouts for heavy vehicles. Drivers should be particularly cautious during monsoon periods for flash flooding and reduced traction.

Traffic Patterns

Traffic analysis along this route suggests moderate congestion at the start and end points, particularly during morning (7 AM - 9 AM) and evening (5 PM - 7 PM) peak hours due to local commuter traffic. Padaiveedu may experience congestion due to commercial activities and local market days. Heavy vehicles may experience delays during these peak times.

Road Quality and Infrastructure

The Salem - Kochi Highway typically has good road quality with regular maintenance. However, interior sections near Puthur may have narrower lanes and occasional patchwork repairs. Signage may be inconsistent in rural segments, requiring attentive driving. Truck drivers should be aware of potential speed bumps and varying road surface conditions.

Alternative Route Suggestions

In case of emergencies, an alternative route via NH 544 through Sankari could be considered. Although longer, it has better established emergency facilities and could be advantageous in adverse weather conditions or road closures on the primary route.

Local Regulations

Transporting hazardous materials through Tamil Nadu involves compliance with local regulations that require documentation and adherence to designated travel hours. There are restrictions against transporting hazardous materials through populated town centers to minimize public risk.

Historical Incidents

There have been occasional minor incidents involving heavy vehicles in this region, typically related to overturned trucks due to sharp turns or slippery conditions during rain. There have been no significant hazardous material incidents reported, but vigilance and adherence to safety protocols are vital.

Environmental and Sensitive Areas

The route passes through mainly rural and agricultural areas. Potential environmental concerns include spills or accidents near water bodies. It’s crucial to avoid contaminating local water supplies or farmlands.

Communication Coverage

Mobile network coverage along this route is generally reliable; however, some rural patches may experience weaker signal strength. Drivers should be prepared for potential brief communication dead zones and plan emergency contacts accordingly.

Emergency Response Times

Emergency response time is estimated to be within 30-45 minutes on major highways like the Salem - Kochi Highway. In more rural sections, response times could be longer, particularly at night or during inclement weather.

Overall Summary of Risk Assessment

This route is relatively safe for transporting hazardous materials under normal conditions, provided the driver maintains vigilance during peak traffic hours, adverse weather conditions, and adheres to local regulations. Key risks include weather-related hazards, traffic congestion, and occasional poor road conditions. Preparedness for communication interruptions and adopting alternative routes in emergencies will enhance overall travel safety. Training drivers on these aspects can significantly minimize risks associated with this journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43968, 77.87345	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45348, 77.85706	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	Medium	11.45697, 77.82719	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.49409, 77.88004	15 KM/Hr
12	Turn	Medium	11.49419, 77.88015	30 KM/Hr
13	Turn	Medium	11.49377, 77.88573	30 KM/Hr
14	Turn	High	11.49363, 77.88590	15 KM/Hr
15	Turn	High	11.49235, 77.89561	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
17	Blind Spot	Blind Spot	11.55839, 78.01288	10 KM/Hr
18	Blind Spot	Blind Spot	11.55800, 78.01305	10 KM/Hr
19	Turn	Medium	11.55679, 78.01024	30 KM/Hr
20	Blind Spot	Blind Spot	11.55501, 78.00596	10 KM/Hr
21	Blind Spot	Blind Spot	11.55486, 78.00606	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45348, 77.85706

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45697, 77.82719

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49363, 77.88590

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49235, 77.89561

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.55800, 78.01305

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55679, 78.01024

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.55501, 78.00596

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.55486, 78.00606